

The Winding Road to *Exascale* and Beyond!

Heavily Drawn from
“Exascale Report: Why so long?
and Are We Really There Yet?”
by Peter Kogge & Bill Dally (Chief Scientist,
Nvidia), PMBS Workshop, SC 2022



What Are We Talking About?

- A Short review of “Supercomputer” Architecture
- HPL: The “Classical” Benchmark
- The Exascale Study and its Predictions
- Rise of New Benchmarks and New Metrics
- The Next Increment: Zetta
- Alternative Architectures: PIM

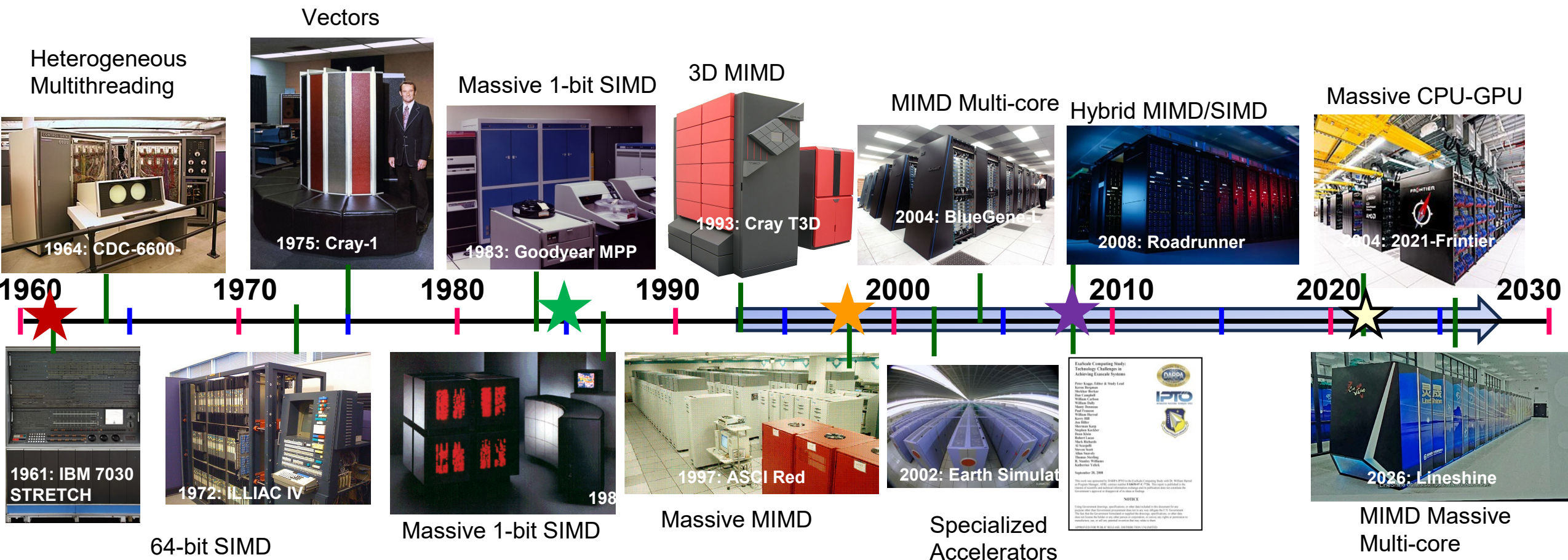
70 Years of “Supercomputers”



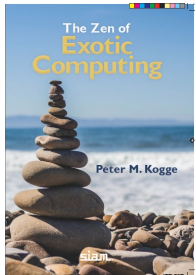
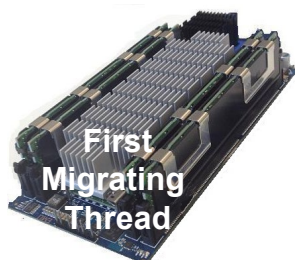
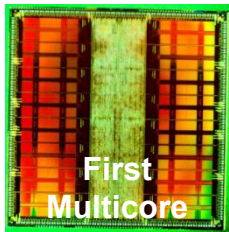
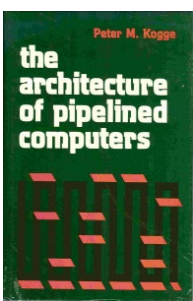
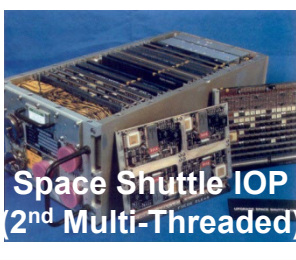
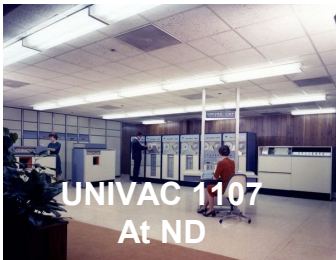
1000X “Faster” Than Prior Star –
for specific HPL benchmark



Supercomputing Conferences:
Formal “Ranking” of TOP500 Computers

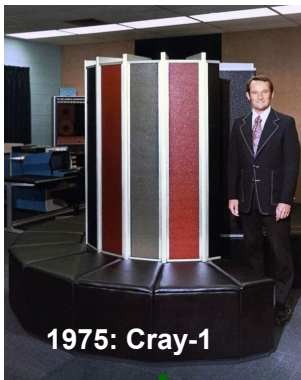


A Personal View



Vectors

Heterogeneous Multithreading



Massive 1-bit SIMD



3D MIMD



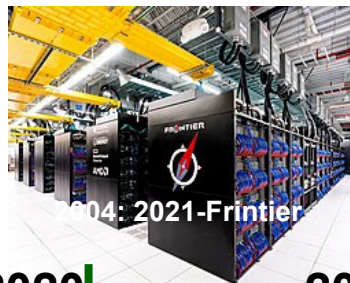
MIMD Multi-core



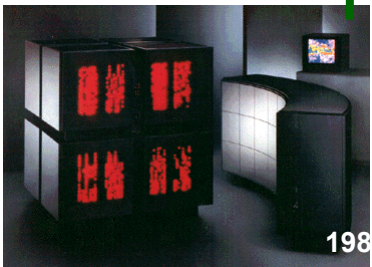
Hybrid MIMD/SIMD



Massive CPU-GPU



64-bit SIMD



Massive 1-bit SIMD



Massive MIMD



Specialized Accelerators



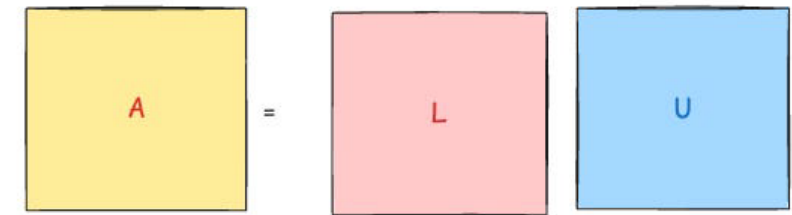
MIMD Massive Multi-core

HPL: Hi Performance Linpack - TOP500.org

- Solve $Ax = b$ in **large dense** systems of linear equations
- Two steps
 - **DGEFA**: Double precision **G**eneral matrix **F**actor.
 - Decompose nxn matrix A into LU
 - $O(n^3)$ flops
 - **DGESL**: Double precision **G**eneral matrix **S**olve.
 - Solve for z in $Lz=b$
 - Solve for x in $Ux = z$
 - $O(n^2)$
- 3 major library routines
 - **DAXPY**: $y = y + \alpha x$, x and y are vectors (90% of ops)
 - **IDAMAX**: locate 1st occurrence of largest absolute value
 - **DSCAL**: scale a vector by a constant

LU decomposition

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$



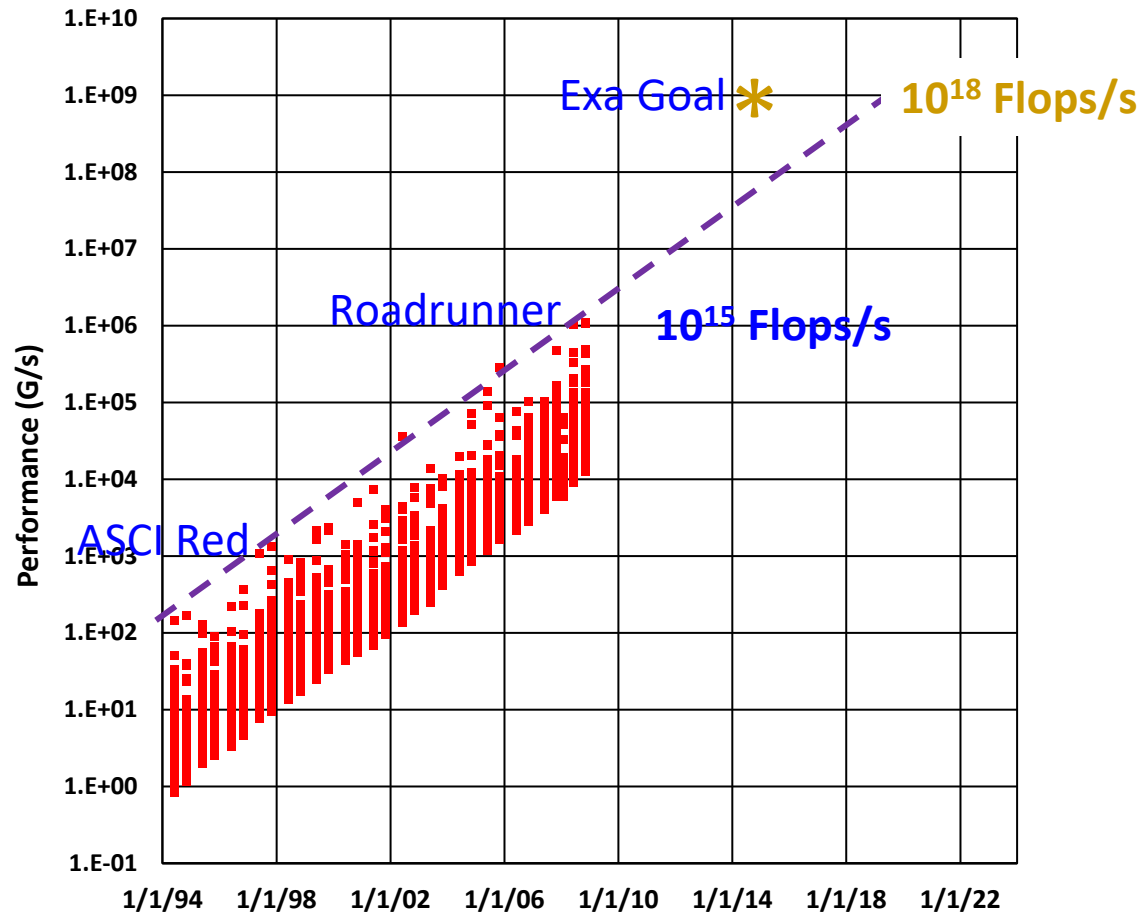
$$A = LU = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

L U

$$L(Ux) = b$$

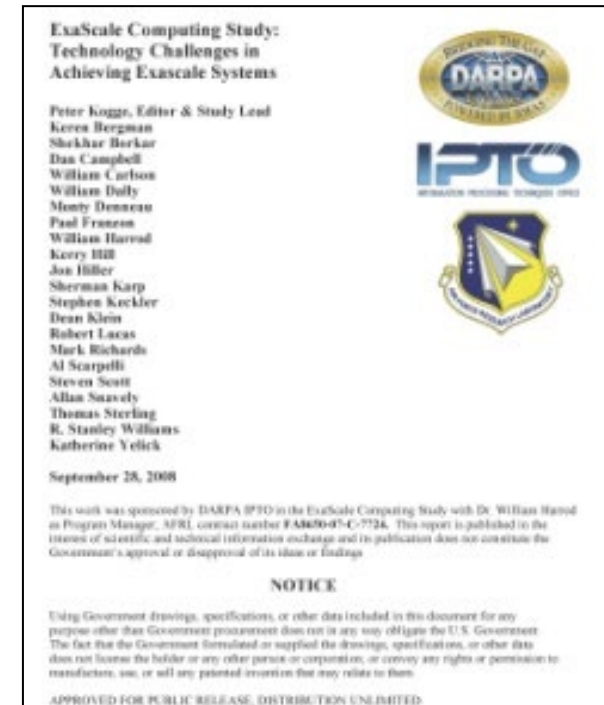
Math Mindset
Tereziya Semenski

The HPL World in 2008

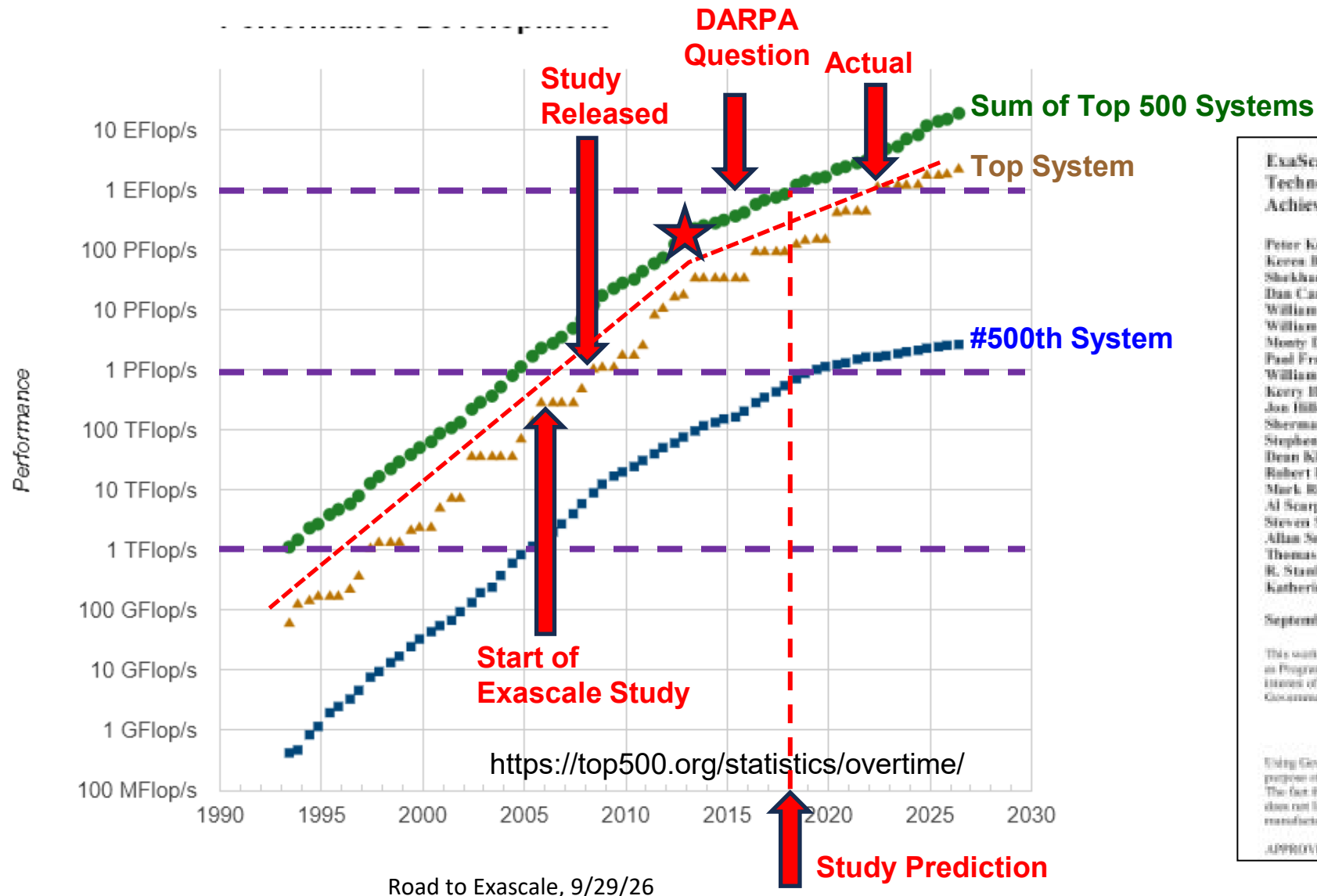


“First Light” for new TOP500 entries

- The Benchmark: HPL
- Roadrunner: 1+ PF/s
- DARPA (Bill Harrod): Exa by 2015?
- 2008 Exascale Report: Yes, but...



Top 500: The Most Referenced “Ranking”



**ExaScale Computing Study:
Technology Challenges in
Achieving Exascale Systems**

Peter Kogge, Editor & Study Lead
 Keen Bergman
 Shikhar Borkar
 Dan Campbell
 William Carlson
 William Dally
 Monty Denneau
 Paul Friesen
 William Harrod
 Kerry Hill
 Jon Hillier
 Sherman Karp
 Stephen Keckler
 Dean Kohn
 Robert Lucas
 Mark Richards
 Al Scarpelli
 Steven Scott
 Allan Savory
 Thomas Sterling
 R. Stanley Williams
 Katherine Yelick

September 28, 2008

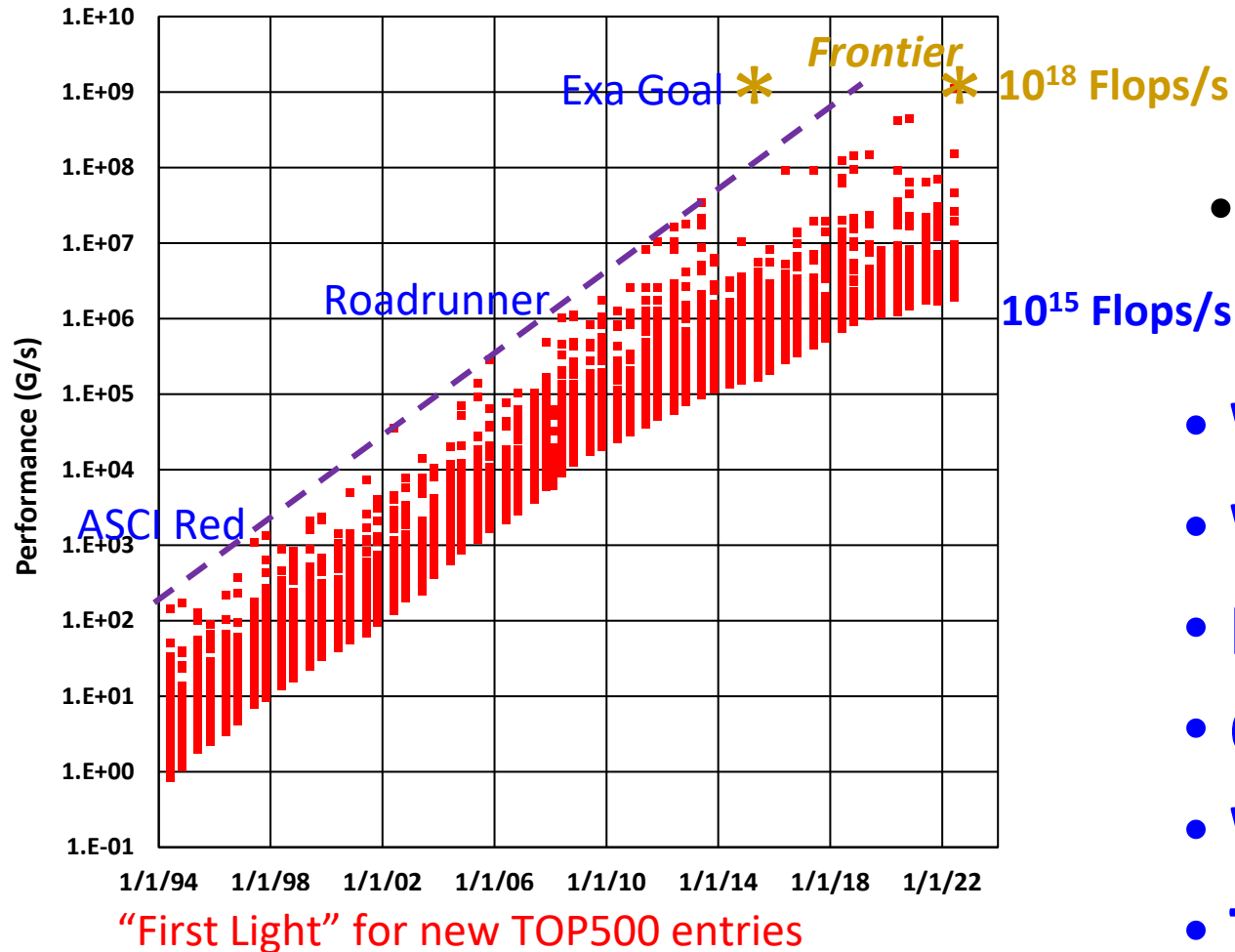
This work was sponsored by DARPA IPTO in the ExaScale Computing Study with Dr. William Harrod as Program Manager. AFRL contract number F49620-07-C-7724. This report is published in the interest of scientific and technical information exchange and its publication does not constitute the Government's approval or disapproval of its ideas or findings.

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The HPL World in 2022



- 2022: Frontier Cracks HPL 1EF/s
 - 7 years after Report Goal
 - 4 years after extrapolating curve
- Bounding Curve Changed in 2013

Obvious Questions

- What Is/Was Exascale?
- What Did 2008 Report Predict?
- More on the Historical Trail
- Comparison to Frontier
- What did Report get Right/Wrong?
- To Zettascale and Beyond

The Exascale Study

- **What *should* “Exascale” Mean?**

- The 2008 state of the art
 - Architectures, Runtimes, Programming, Metrics
- 2008 Application Characteristics
 - Computation vs Memory intensive Apps, Scaling, Concurrency
- Technology Roadmaps
 - Logic: Silicon and Non, Memory, Storage, Interconnect, Packaging, Resiliency, Programming Models

- **Strawman Designs**

- Subsystem projections, Evolutionary designs (Heavy and lightweight), Aggressive design

- **Challenges & Research Areas**

- Power, power, power, & power
 - Memory capacity & bandwidth
 - Programmability
 - Reliability
- } **Remain**
- } **Practically Solved**

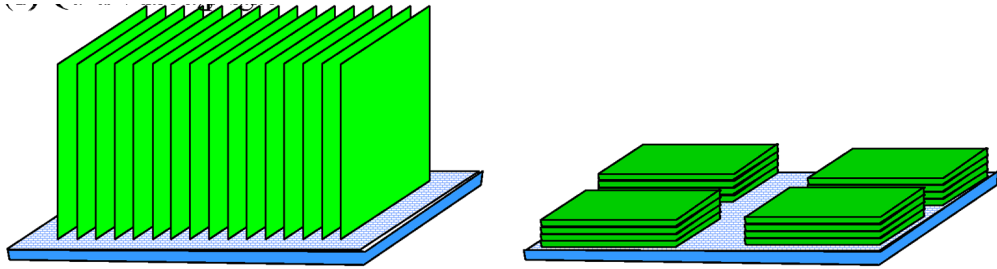
What Was/Is Exascale?

- Report Emphasis: *Try* to change focus from flops
- Goal: **overall** 1000x capability over “Petascale” by 2015
 - In Same Footprint for Supercomputer at max 20MW
 - 1000X in a rack (peta scale)
 - 1000X in a module (tera scale)
- Not just flops but
 - Memory
 - Memory Bandwidth
 - Network Bandwidth
 - ...
- Plus ability to program massive concurrency

Technologies Investigated

- Logic: power, area, energy, clock
 - CMOS: hi perf/low voltage
 - Options: hybrid, superconducting
 - Voltage scaling
- Main Memory
 - SRAM, DRAM, NAND, Alternatives
 - Reliability, packaging, power
- Storage Memory
 - Disk, Holographical, Archival
- Interconnect: esp. energy
 - On chip
 - DRAM to Processor (Stacking)
 - Intra/inter module
 - Rack to rack
 - Electrical vs optical
- Packaging and Cooling
- Resiliency & Checkpointing
- Programming Models

2015 Aggressive Strawman Design (2013 Tech)



Node: 742 simple cores/chip with 4 FPUs @ 1.5GHz

- 32nm CMOS with 30Gb/s SERDES
- 16 Memory channels: each 1 GB *Stacked* DRAM
- 150 Watts w/o routing chip

Group: 12 nodes with 12 64-radix router chips

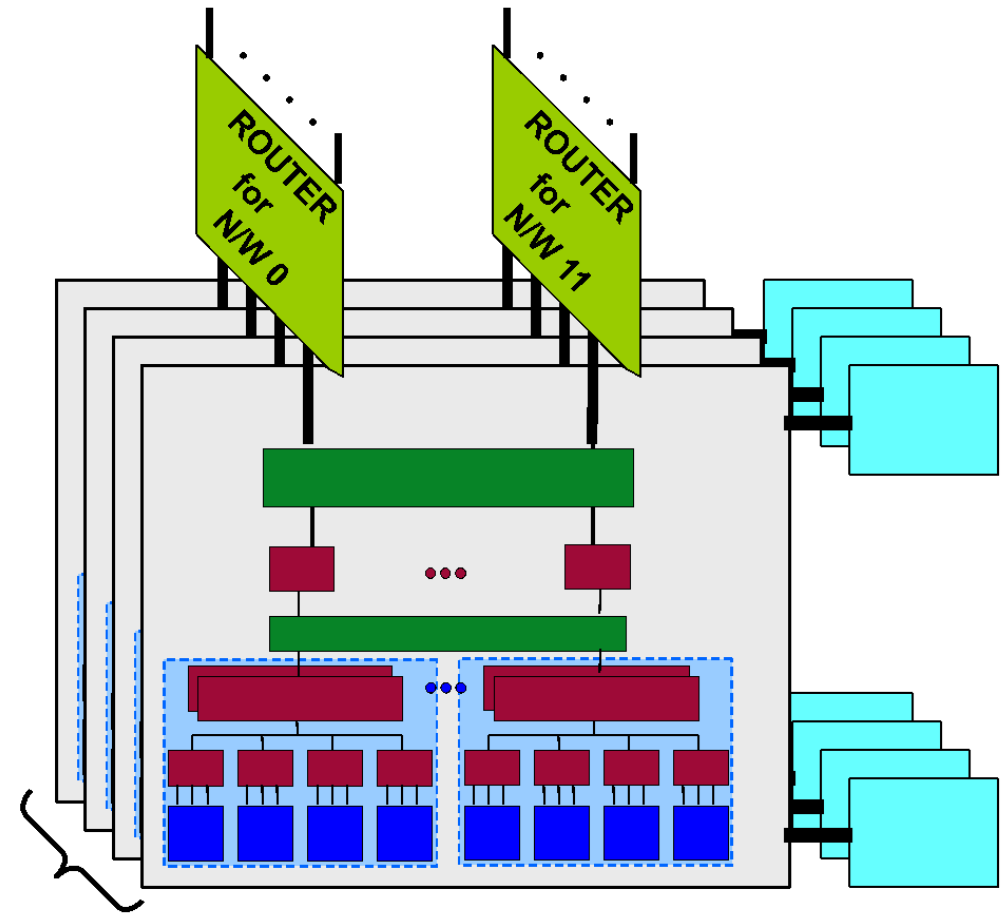
- Includes 16 12GB SATA drives for checkpointing

Cabinet: 32 Groups = 384 nodes

- Assumed max power of 120KW

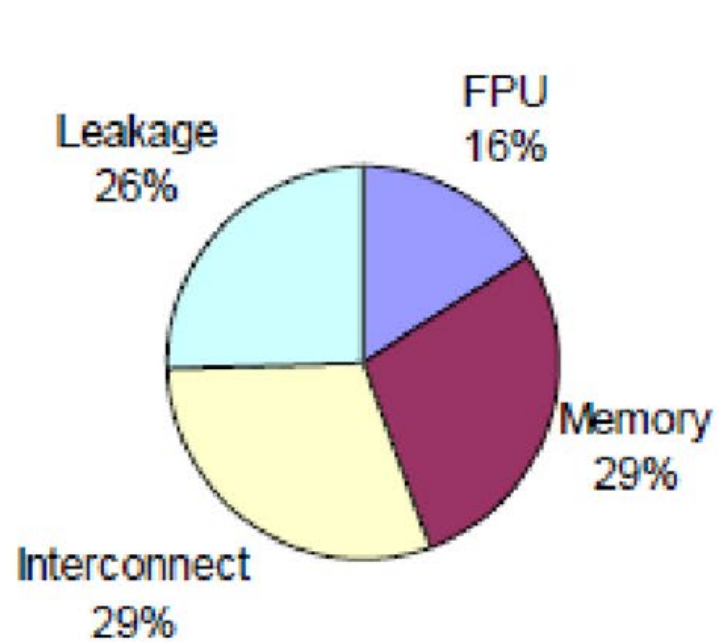
System: 583 Cabinets, 67MW

- 3-hop Dragonfly interconnect
- 166 million cores with 664 million FPUs

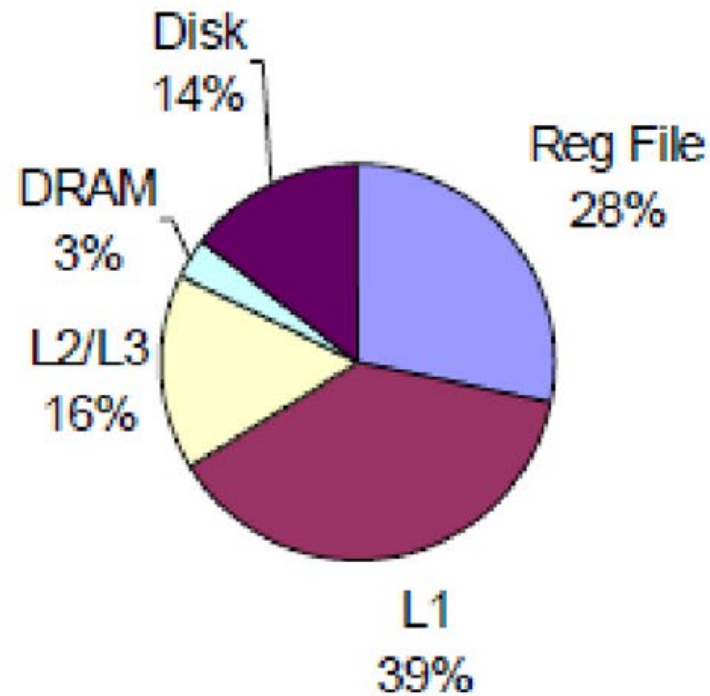


Est. 14.9 GF/W

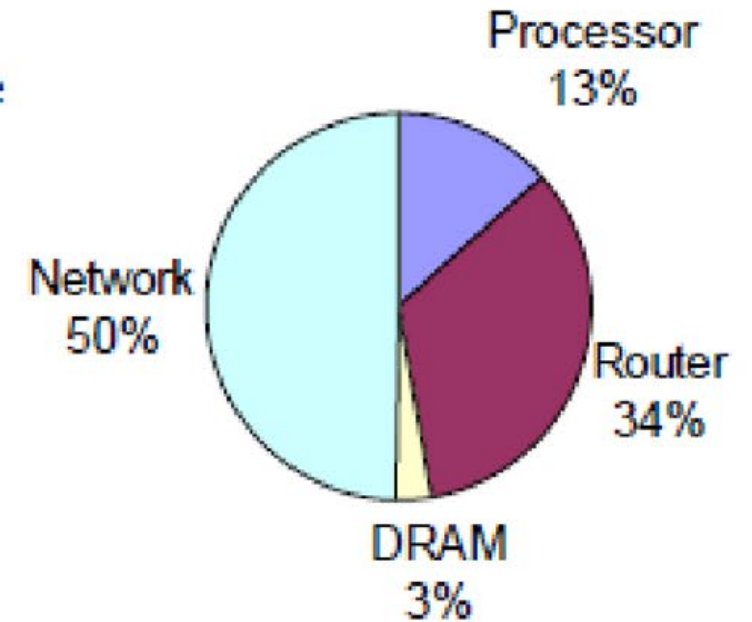
Where Did the Energy Go?



(a) Overall System Power



(b) Memory Power



(c) Interconnect Power

2018: Summit – An Exascale “Could Have Been”

- **Nodes:**

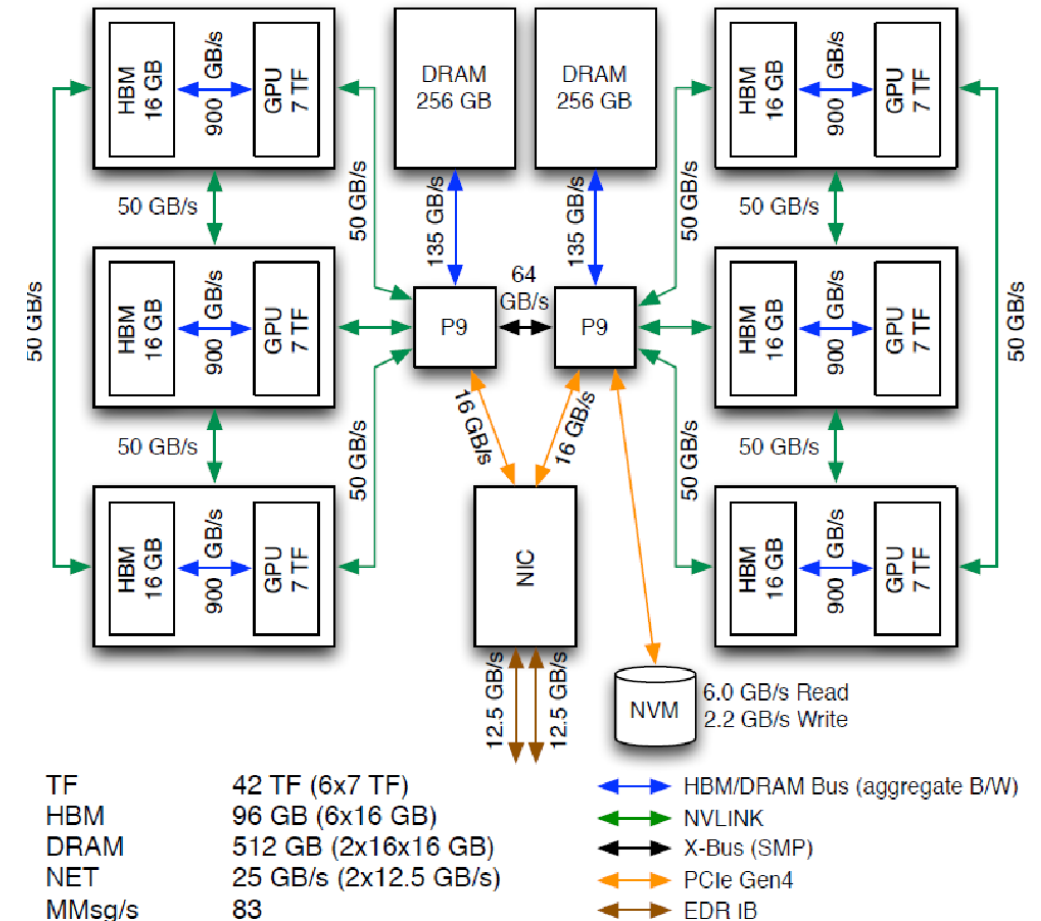
- Dual 22 core Power 9
- Hex NVIDIA GV100
- Mixed DRAM/HBM (Stacked)

- **Cabinet:** 18 Nodes, 55KW

- **System:** 256 compute, 9.8 MW

Interesting Observation: 6.7X expansion of Summit

- ~1+ EF/s sustained
- At about 67 MW!



HBM & DRAM speeds are aggregate (Read+Write).

All other speeds (X-Bus, NVLink, PCIe, IB) are bi-directional.

Summit System Overview, T. Papatheodore,

6/1/18

Measured 14.7 GF/W

Strawman vs. Summit

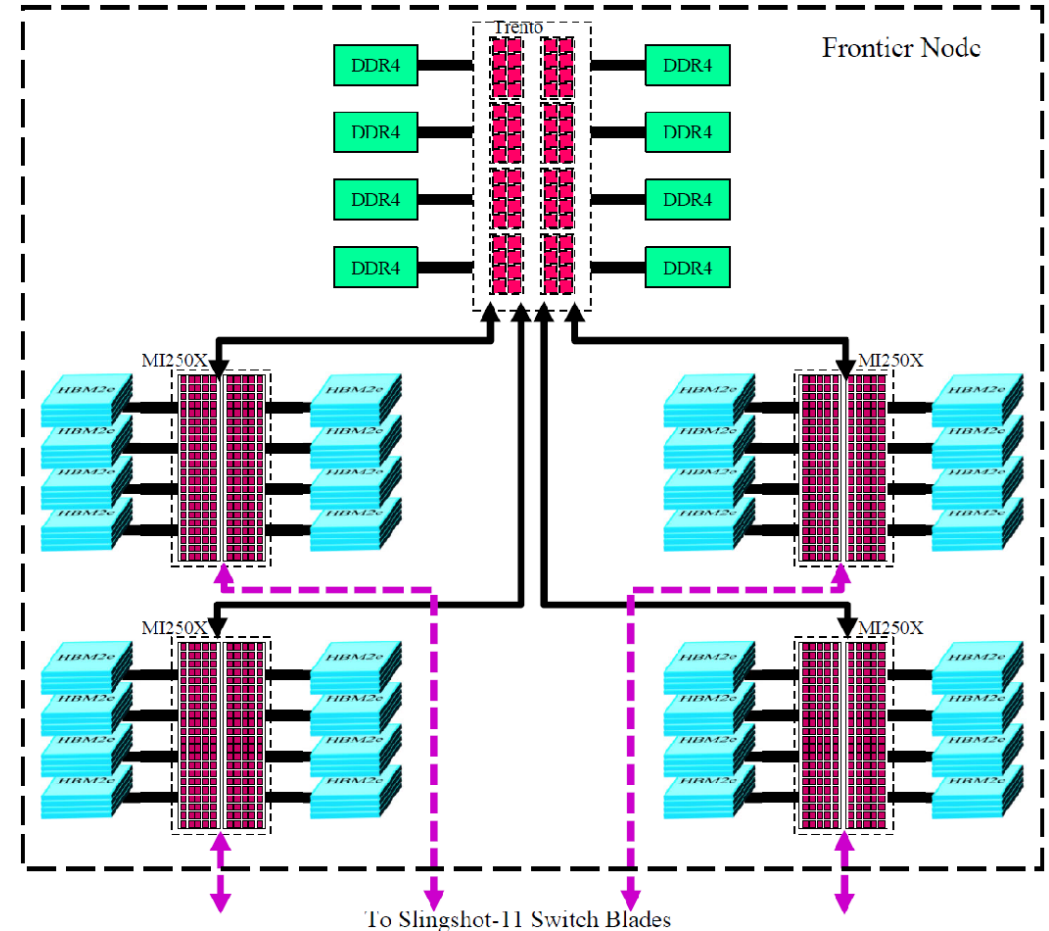
	RR	Strawman	Summit
Year	6/2008	2015	11/2018
Best Tech	65nm	32nm	16nm
Peak (PF/s)	1.38	2,000	201
Sustained (PF/s)	1.04	1,000	148
Power (MW)	2.35	67.7	9.8
Efficiency (GF/W)	0.44	14.9	14.7
Memory (PB)	0.04	3.5	2.8
Bandwidth/flop (B/F)	0.28	0.08	0.13
Mem BW (PB/s)	0.38	158	27
Bisection(TB/s)	0.192	210	105
FPU's (M)	0.464	664	144
Cabinets	296	583	256
Floorspace (m^2)	557	1195	520

*Summit: Could have matched
Strawman if scaled up ~6.7X*

63% better

2022 Frontier Node

- Heterogeneous Processors
 - 64-core 2GHz CPUs
 - Quad GPUs: closer to Strawman
 - But more FPU/core
 - And slightly faster
- Chiplet design
- Mixed memory hierarchy
 - 8 DDR4 DRAM Channels
 - 8 HBM2e stacks/GPU
- Quad network ports



Measured 52.2 GF/W

2022 Frontier System

- **Blade:** 2 nodes
- **Chassis:** 8 Processor Blades
 - With up to 8 Router Blade
 - Arranged perpendicularly
- **Cabinet:** 8 Chassis = 128 nodes
 - Water cooled up to 400KW
 - Over 2X footprint of Strawman
- **System:** 74 compute cabinets
 - With *additional* Cooling Units
 - Again Dragonfly topology



By OLCF at ORNL - <https://www.flickr.com/photos/olcf/52117623843/>, CC BY 2.0, <https://commons.wikimedia.org/w/index.php?curid=119231238>

More Detailed Comparison

	RR	Strawman	Summit	Frontier
Year	6/2008	2015	11/2018	6/2022
Best Tech	65nm	32nm	16nm	6nm
Peak (PF/s)	1.38	2,000	201	1,686
Sustained (PF/s)	1.04	1,000	148	1102
Power (MW)	2.35	67.7	9.8	21.1
Efficiency (GF/W)	0.44	14.9	14.7	52.2
Memory (PB)	0.04	3.5	2.8	9.4
Bandwidth/flop (B/F)	0.28	0.08	0.13	0.07
Mem BW (PB/s)	0.38	158	27	125
Bisection(TB/s)	0.192	210	105	540
FPU's (M)	0.464	664	144	534
Cabinets	296	583	256	74
Floorspace (m^2)	557	1195	520	678

*Summit: Could have matched
Strawman if scaled up ~6.7X*

*Frontier: not even close to
1000X over Roadrunner in
other categories*

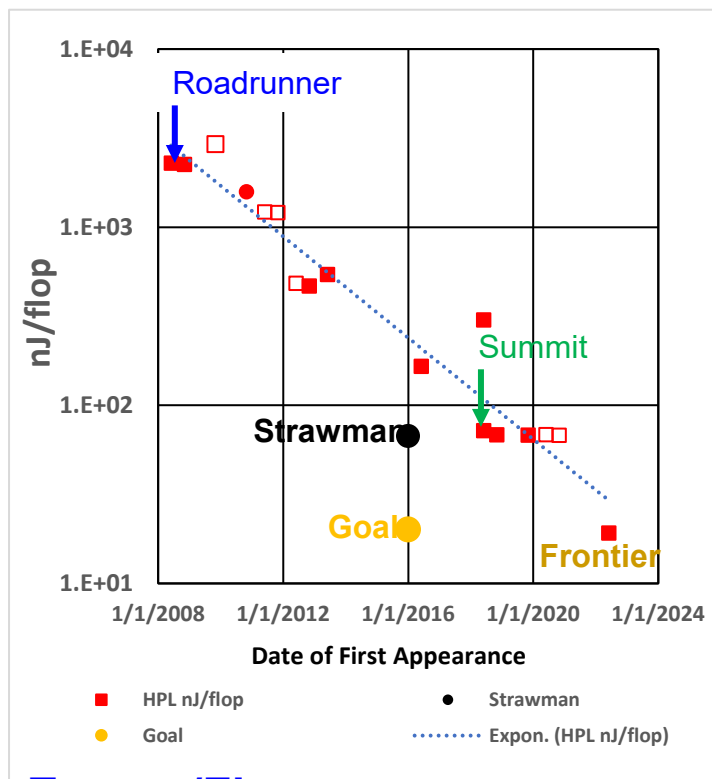
Technology Changes from Then to Now/2022

	Roadrunner	Strawman	Summit	Frontier
Gate Length (nm)	65	32	16	6
Metal 1 pitch (nm)	180	100	64	40
Energy ⁻¹	1	1.8	2.8	4.5
Area ⁻¹	1	3.2	7.9	20.3

Also nowhere near
Dennard Scaling:
Voltage scaling has
stopped

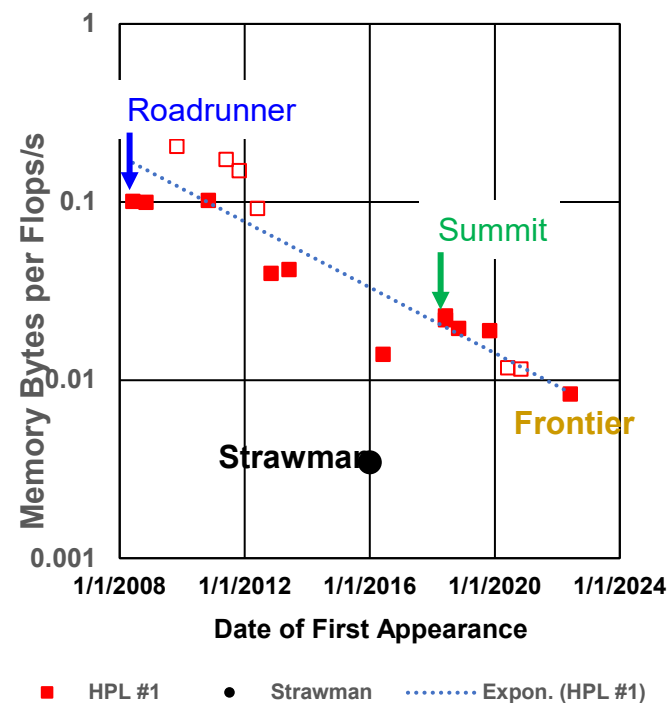
Nowhere near 100X:
Metal 1 pitch dominates density:
Not transistors

Changes in System Characteristics



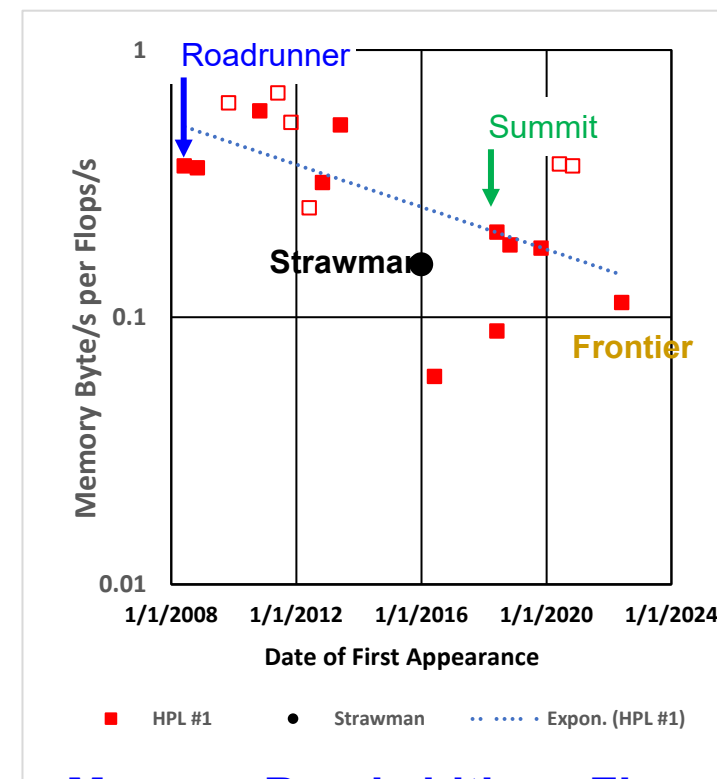
Energy/Flop:

- Declined >100X since 2008
- Summit matched Strawman in 2018



Memory Capacity vs Flops/s:

- Declined >10X since 2008
- Strawman was even worse



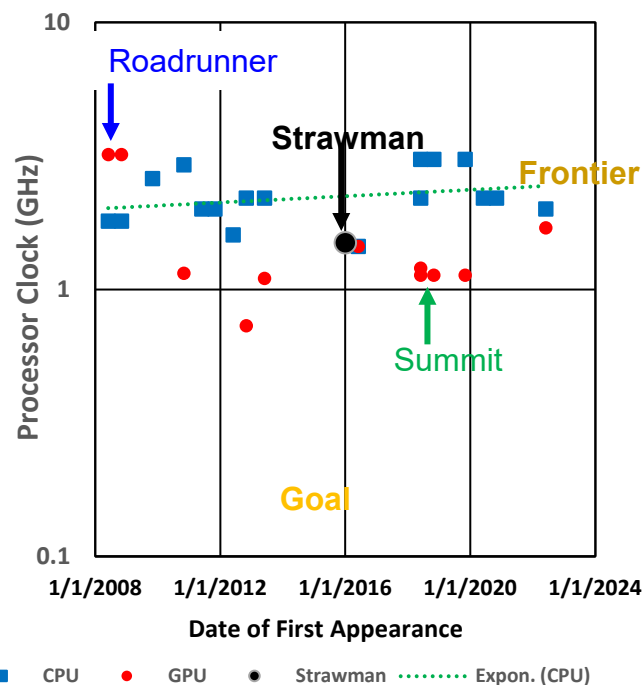
Memory Bandwidth vs Flops/s:

- Declined >3X since 2008
- Strawman was down 2X

■ Heterogeneous Processor Architecture

□ Homogeneous Processor Architecture

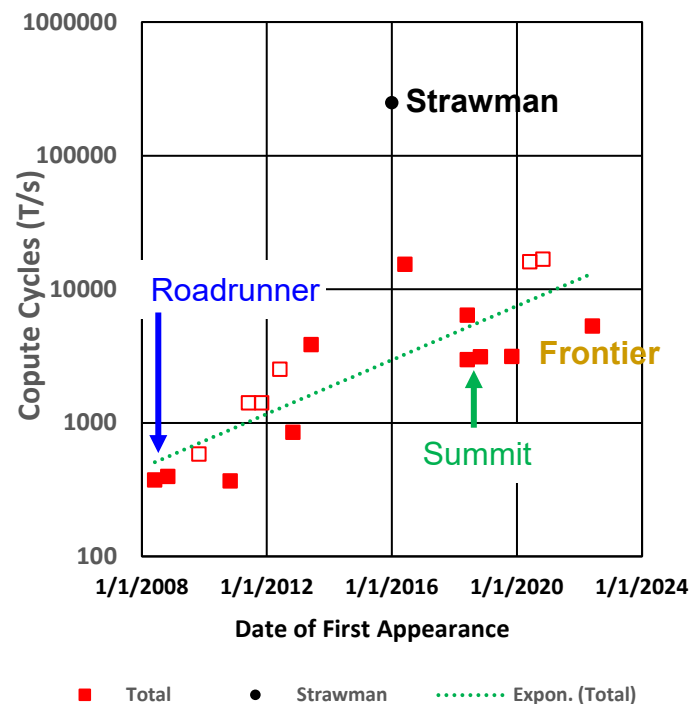
Changes in Architecture Characteristics



Processor Clock:

- Essentially Flat since 2008
- GPUs ran slower than CPUs

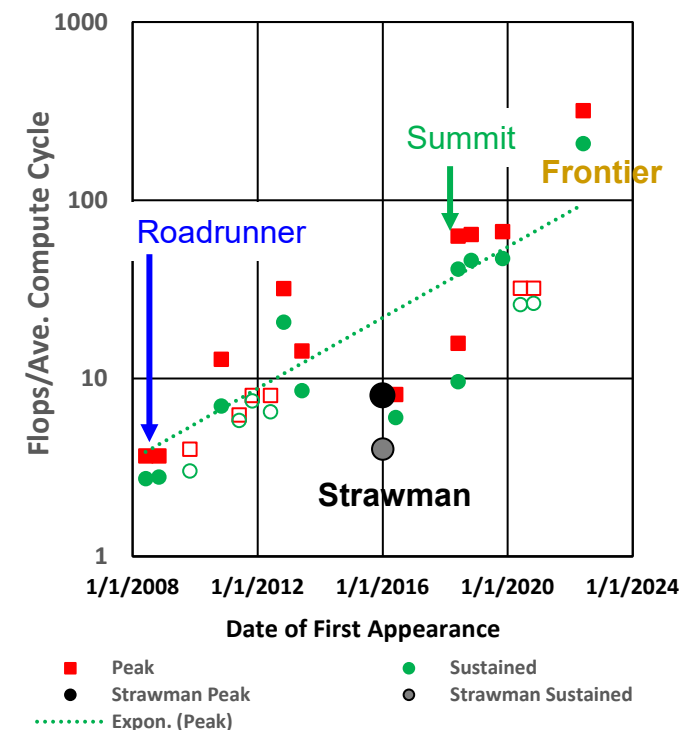
■ Heterogeneous Processor Architecture



Aggregate Compute Cycles:

- Increased >14X since 2008
- Strawman had huge # of cores
 - But only 4 FPU wide each

□ Homogeneous Processor Architecture



Flops per Cycle:

- Exploded with Advent of GPUs
- Strawman didn't go far enough

Frontier vs Strawman

	Road-Runner	2008 Strawman	Frontier
System Counts			
Nodes/Blade	1	12	2
Blades/Chassis	4	1	8
Chassis/Cabinet	3	32	8
Nodes/Cabinet	12	384	128
Total Nodes	3060	223,872	9,408
Cores/Node	40	742	944
MACs/Node	76	2,968	56,832
Total MACs	232K	665M	535M
Memory Metrics			
Total Memory (TB)	36	3,498	9,408
Total Memory BW (TB/s)	378	157,605	125,239
Network Bandwidth Metrics			
Network ports/node	1	12	4
Total Network ports	3,060	2.7M	37,632
Switch Chips/Cabinet		384	64*
Switch Radix	24	64	64
Total Switch Chips	900	223,872	4,736*
Signal Rates (Gb/s)	4	30	56
Inj. B/W/Node (GB/s)	2	180	100
Bisection B/W (TB/s)	0.192	210	540
* Assuming 8 switch cards/chassis			

- Strawman's **huge #s of nodes**
 - Exploded # of Network ports
 - And thus huge switching costs
- Frontier had fewer, **bigger nodes**
 - Reduced network ports
- **Comparable** Memory Bandwidth
 - Use of wide stacked memory
 - But only 3X capacity
- Essentially **same N/W topology**
 - But 2X better SERDES
 - And 2+X better bisection B/W

Frontier vs Roadrunner: Did We Get 1000X?

	Road-Runner	Frontier	Growth Ratio
GFlops/s/core	8.4	126	15
GFlops/s/chip	56	23,426	419
TFlops/s/node	0.34	117	349
TFlops/s/cabinet	4	14,993	3,726
TFlops/s/sq. ft.	0.17	151	882
Flops/core/cycle	2.74	208	75
Flops/cycle ¹	3.2E5	6.7E8	2,022
Flops/Mem byte	9.9	119	12.1
Flops/Mem BW byte	2.7	8.8	3.25
Flops/Inj. byte	168	1,171	7
GFlops/watt	0.44	52.2	119
Watts/core	19.24	2.4	1/8
Watts/chip	128	449	3.5
Watts/node	766	2,243	2.9
All cores and all chips included			
¹ Using clock for major compute core.			

- Flops/s **exceeded** 1000X / cabinet
 - But huge cabinets
 - Within 3X for chip & node
- **>100X** in flops/s per watt
 - And flops/cycle
- **Miserable increase** in Memory, Memory Bandwidth. N/W Injection Bandwidth

Report Card

What We Got Right

- CMOS, flat clocks
- Large # of wide simple cores
- Aggressive memory hierarchy
- Stacked memory
- Near reticle-limited dies
- Energy of movement predominates
- Near billion-way concurrency
- Memory concerns were valid
- Dragonfly with hi radix switches
- N/W signaling rate would improve

What We Missed

- Heterogeneous designs
- SIMD width much larger
- Stacked memory: more ports/lower transfer rate
- Machine Learning & short FP
- Massive 500W chips coolable
- Reliability not a show-stopper
- New programming models

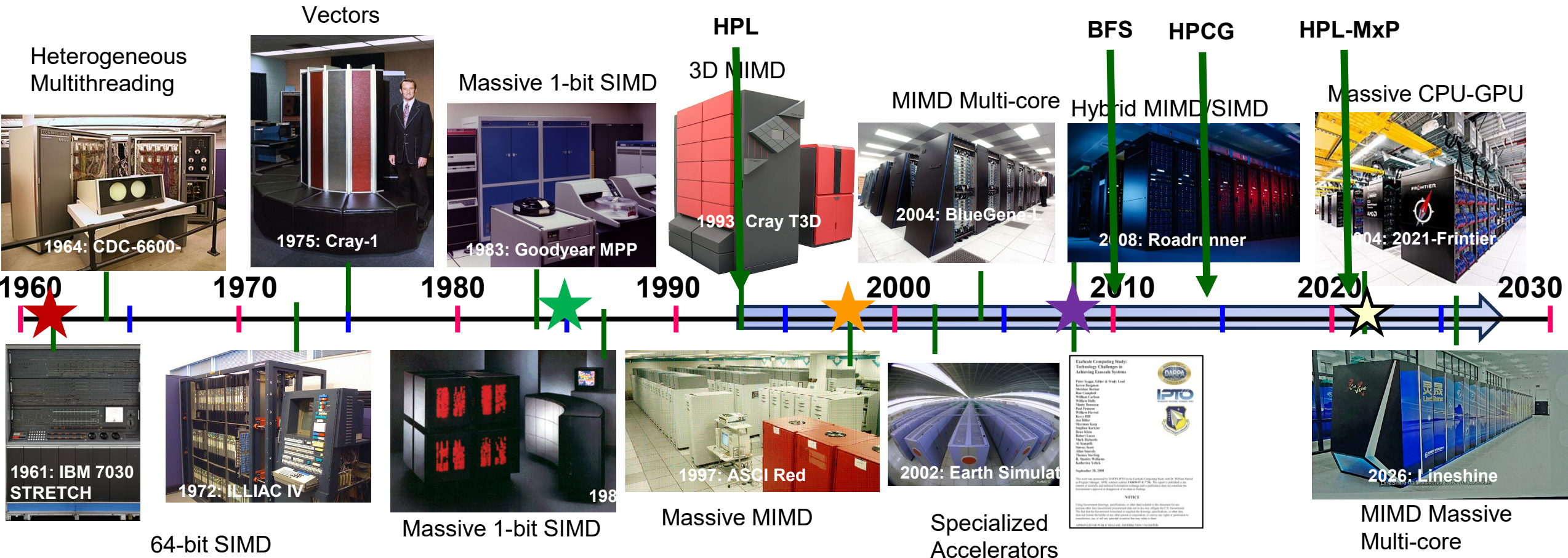
After 20+ Years Was/Is HPL the Right Reference?

- Power now primary!
 - New metric **Flops/Watt** – **GREEN 500**
- Real World Apps are NOT Dense Matrices!
 - New benchmark: **BFS** graph processing
 - New benchmark: **HPCG** Sparse 3D problems
- Rise of AI introduced very small floating point formats
 - Huge reduction in memory and memory bandwidth needs
 - New benchmark: **HPL-MxP** HPL/HPCG-like in short precision
- All above have “Green” metrics

70 Years of “Supercomputers”

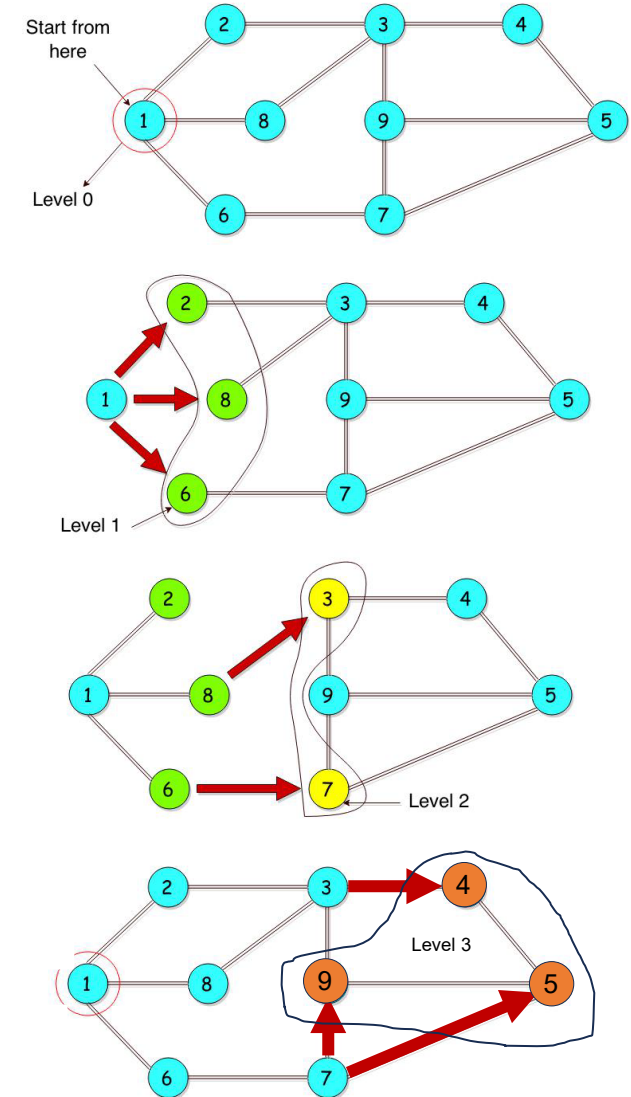
★ 1000X “Faster” Than Prior Star

Supercomputing Conferences:
Formal “Ranking” of TOP500 Computers



BFS: Breadth First Search – GRAPH500.org

- Developed to stress non-flop operations
 - By ND Quad-domer
- Find spanning tree for large graphs $O(|E|)$
 - Start at random vertex as “level 0”
 - Mark all non-visited neighbors as “level 1”
 - Repeat from level k to level k+1
- Metric: Execution time/# edges
- **Beamer's Algorithm**: once frontier is large
 - Reverse direction and see if non-visited vertices have edge to any visited



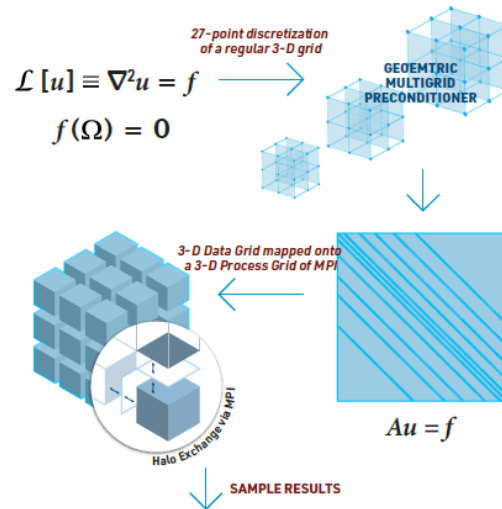
HPCG: High Performance Conjugate Gradient (HPCG.org)



NOVEMBER 2021 TOP 5				
		SITE	COUNTRY	Rmax FLOPS
1	Fugaku	RIKEN	Japan	442
2	Summit	DOE/SC/ORNL	USA	149
3	Perlmutter	DOE/SC/LBNL	USA	65
4	Sierra	DOE/NNSA/LLNL	USA	95
5	Selene	NVIDIA	USA	64

The HPC Conjugate Gradient (HPCG) benchmark uses a preconditioned conjugate gradient (PCG) algorithm to measure the performance of HPC platforms with respect to frequently observed, yet challenging, patterns of execution, memory access, and global communication.

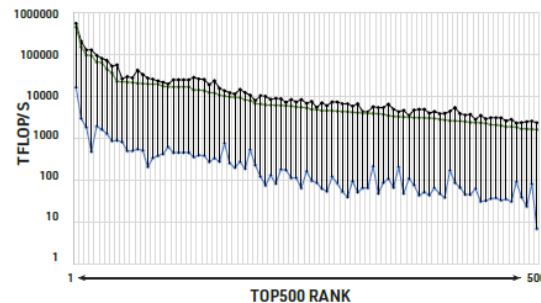
The PCG implementation uses a regular 27-point stencil discretization in 3 dimensions of an elliptic partial differential equation (PDE) with zero Dirichlet boundary condition. The 3-D domain is scaled to fill a 3-D virtual process grid of all available MPI process ranks. The CG iteration includes a local and symmetric Gauss-Seidel preconditioner, which computes a forward and a back solve with a triangular matrix. All of these features combined allow HPCG to deliver a more accurate performance metric for modern HPC hardware architectures.



PRECONDITIONED CONJUGATE GRADIENT SOLVER

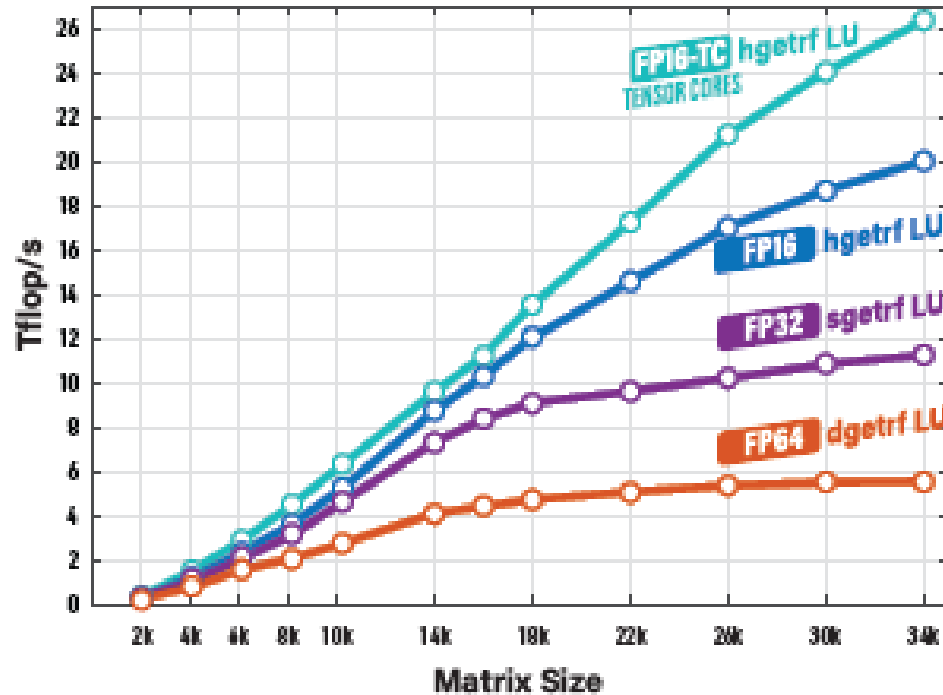
```

p0 ← x0, r0 ← b - Ap0
for i = 1, 2, to [max.iterations] do
  z1 ← M⁻¹rᵢ₋₁
  if i = 1 then
    p1 ← z1
    α1 ← dot.prod(rᵢ₋₁, z1)
  else
    α1 ← dot.prod(rᵢ₋₁, z1)
    β1 ← α1/αᵢ₋₁
    p1 ← β1pᵢ₋₁ + z1
  end if
  α1 ← dot.prod(rᵢ₋₁, z1)/dot.prod(p1, Ap1)
  xᵢ₊₁ ← xᵢ + α1p1
  r1 ← rᵢ₋₁ - α1Ap1
  if ||r1||₂ < [tolerance] then
    STOP
  end if
end for
    
```



- More realistic than HPL
 - Solve 3D PDE using 3x3x3 cube around each point
 - Conjugate gradient iteration
- Each MPI rank gets a sub-cube
 - Represented as a banded matrix
- Each iteration shares surfaces
- Major computation: SPMV product
 - Memory bound on systems designed for HPL
- Flop rate a small fraction of HPL

HPL-MXP: Mixed Precision (HPL-MxP.org)

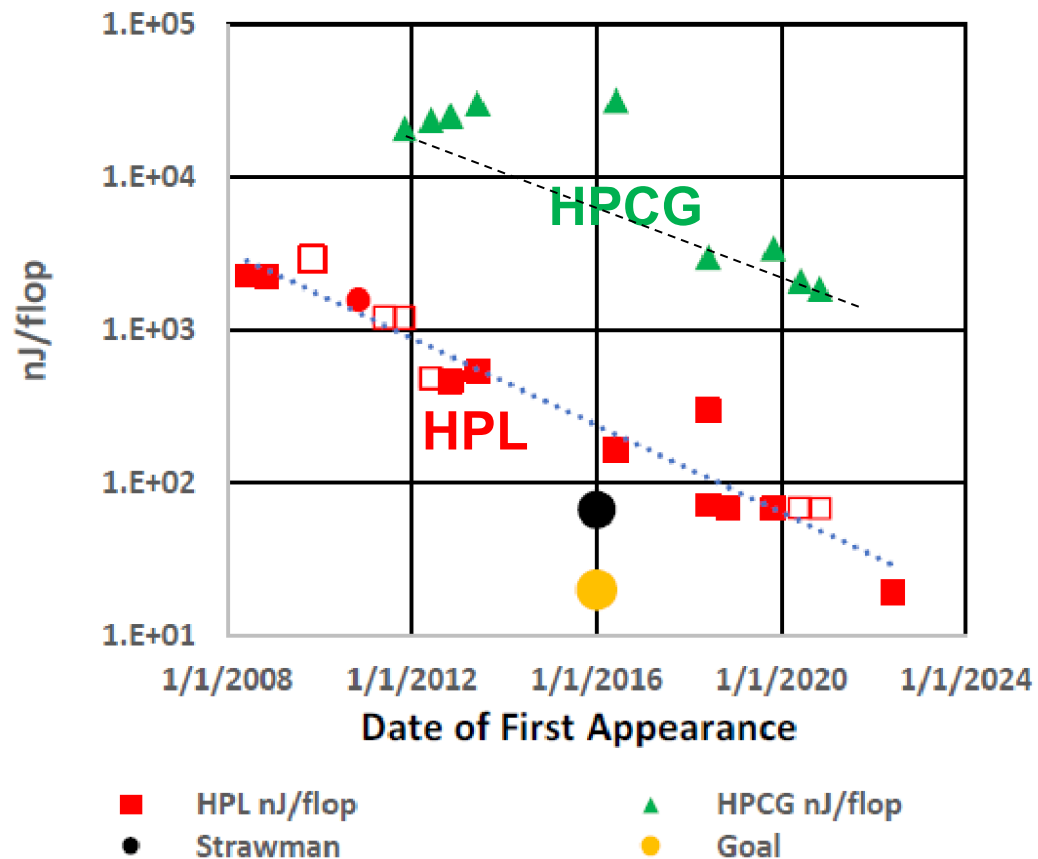


- Recognize emergence of <<64b FP (down to 4b)
- Solve $Ax = b$ as in HPL
 - Via LU decomposition
 - But in lower than 64b quantization
 - With iterative refinement to restore accuracy

HPL-MxP MIXED-PRECISION BENCHMARK JUNE 2025

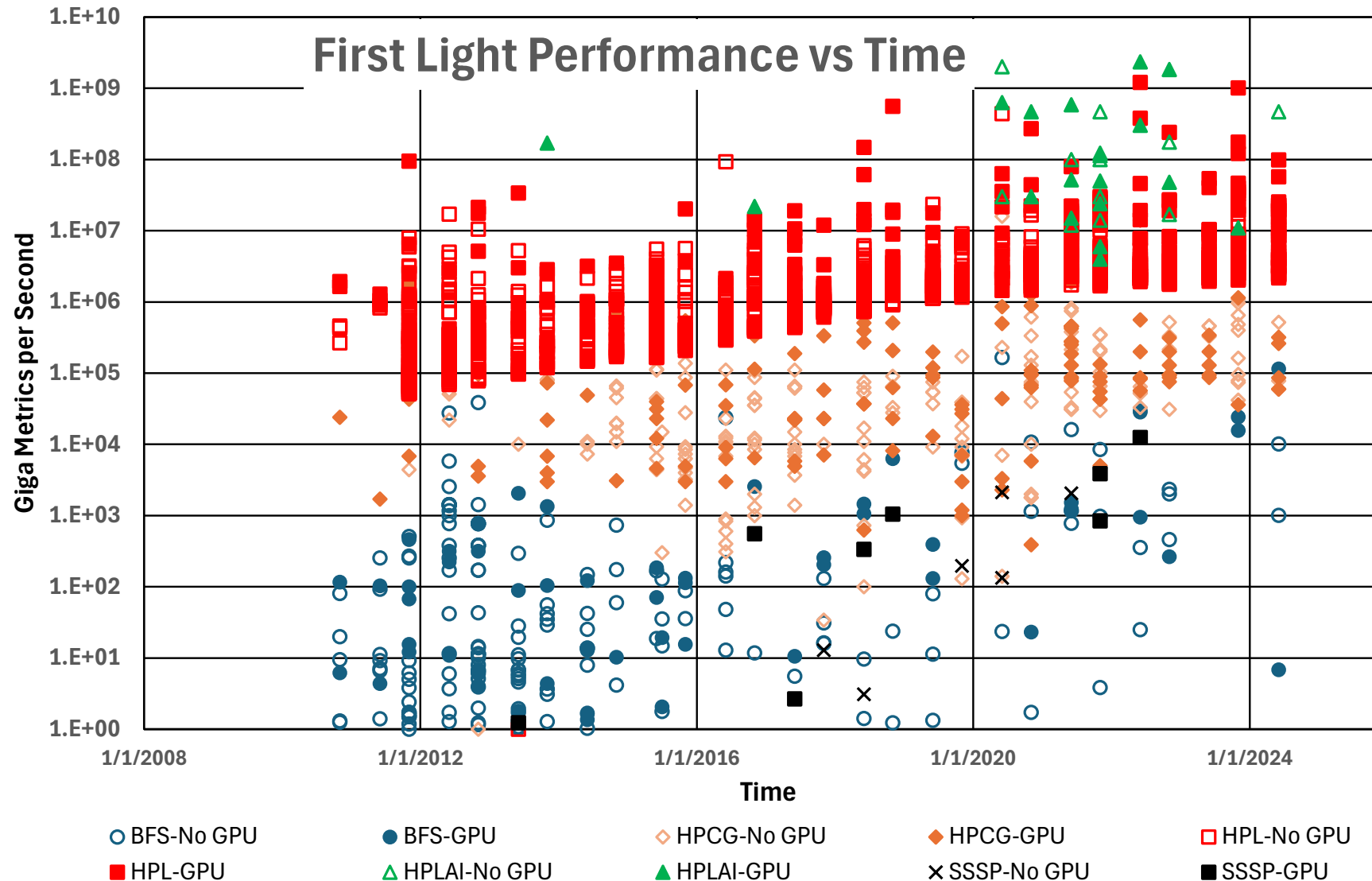
			CORES	HPL-MxP EFLOP/S	TOP 500 RANK	HPL RMxP EFLOP/S	SPEEDUP
1	EL CAPITAN	HPE Cray EX255a, AMD 4th Gen EPYC 24C 1.8GHz, AMD Instinct MI300A, Slingshot-11, TOS	11,039,616	16.680	1	1.7420	9.6
2	AURORA	HPE Cray EX, Intel Exscale Compute Blade, Xeon CPU Max 9470 62C, 2.4 GHz, Intel GPU MAX, Slingshot-11	8,159,232	11.643	3	1.0120	11.5
3	FRONTIER	HPE Cray EX235a, AMD Optimized 3rd Gen EPYC 64C 26Hz, AMD Instinct MI250X, Slingshot-11	8,560,640	11.390	2	1.3530	8.4
4	ABCI 3.0	HPE Cray XD670, Xeon Platinum 8558 48C 2.1GHz, NVIDIA H200 SXM5 141 GB, Infiniband NDR200	479,232	2.363	15	0.1451	16.3
5	LUMI	HPE Cray EX235a, AMD Optimized 3rd Gen EPYC 64C 26Hz, AMD Instinct MI250X, Slingshot-11	2,752,704	2.350	9	0.3797	6.2
6	FUGAKU	Fujitsu A64FX 48C 2.2GHz, Tofu D	7,630,848	2.000	7	0.4420	4.5
7	LEONARDO	BullSequana XH2000, Xeon Platinum 8358 32C 2.6GHz, NVIDIA A100 SXM4 64 GB, Quad-rail NVIDIA HDR100 Infiniband	1,824,768	1.842	10	0.2412	7.6
8	TSUBAME4	HPE Cray XD665, AMD EPYC 9654 96C 3.55GHz, NVIDIA H100 SXM5 94Gb, Infiniband NDR200	172,800	0.641	46	0.0396	16.2
9	SELENE	NVIDIA DGX A100, AMD EPYC 7742 64C 2.25GHz, NVIDIA A100, Mellanox HDR Infiniband	555,520	0.630	30	0.0635	9.9
10	PERLMUTTER	HPE Cray EX235n, AMD EPYC 7763 64C 2.45GHz, NVIDIA A100 SXM4 40 GB, Slingshot-11	888,832	0.590	25	0.0792	7.4

HPCG: Same App as HPL but Sparse Data



- Far less energy efficient
 - H/W resources underutilized
 - Insufficient memory B/W
 - Need 8-10 memory bytes/flop
 - Rate of improvement not as much
- Clearly “Flops at all costs” *not* long term general solution**

The Multi-Faceted World of 2020s



- “First Light”: 1st appearance
- Color=Benchmark
 - HPL
 - HPCG
 - HPL_MxP
 - BFS
 - SSSP
- Fill
 - Solid: with GPUs
 - Hollow: without

“Zettascale” in 2036?

- Zettascale HPL (10^{21} flops/s) not feasible
 - 64bit FPU might go from today's 10pJ to 2-3pJ
 - Just math path of ZettaFLOPS HPL machine would consume 2-3GW
- Better: 1000X for today's critical apps *in same footprint*
 - Multi-physics esp. Climate modeling; Molecular dynamics; ...
 - Machine Learning; Bioinformatics; ...
- Non-starter: Technology scaling
 - Effective gate lengths may drop 3+X to 1-2nm
 - But metal pitch unlikely to improve significantly
 - 3D stacking might give 8X, but costly & little energy improvement

Bridges to Zettascale

- Efficiency via Specialization
 - Reduced precision & specialized data types & operators
 - Memory system specialized to minimize data movement
 - E.g. 15,000X for bioinformatics accelerator
- Reduce design costs via chiplets
 - Design just the accelerator core, not the whole system
- Growth of AI into Scientific Computation
 - Orders of magnitude improvement on some problems
- Explicit Support for Sparsity
 - Fine grain memory to avoid overfetch
 - Finer-grained transfer on networks for better small-message traffic
 - Efficient scatter/gather, pointer walkers

Data Centers – Are We Already at Zetta?



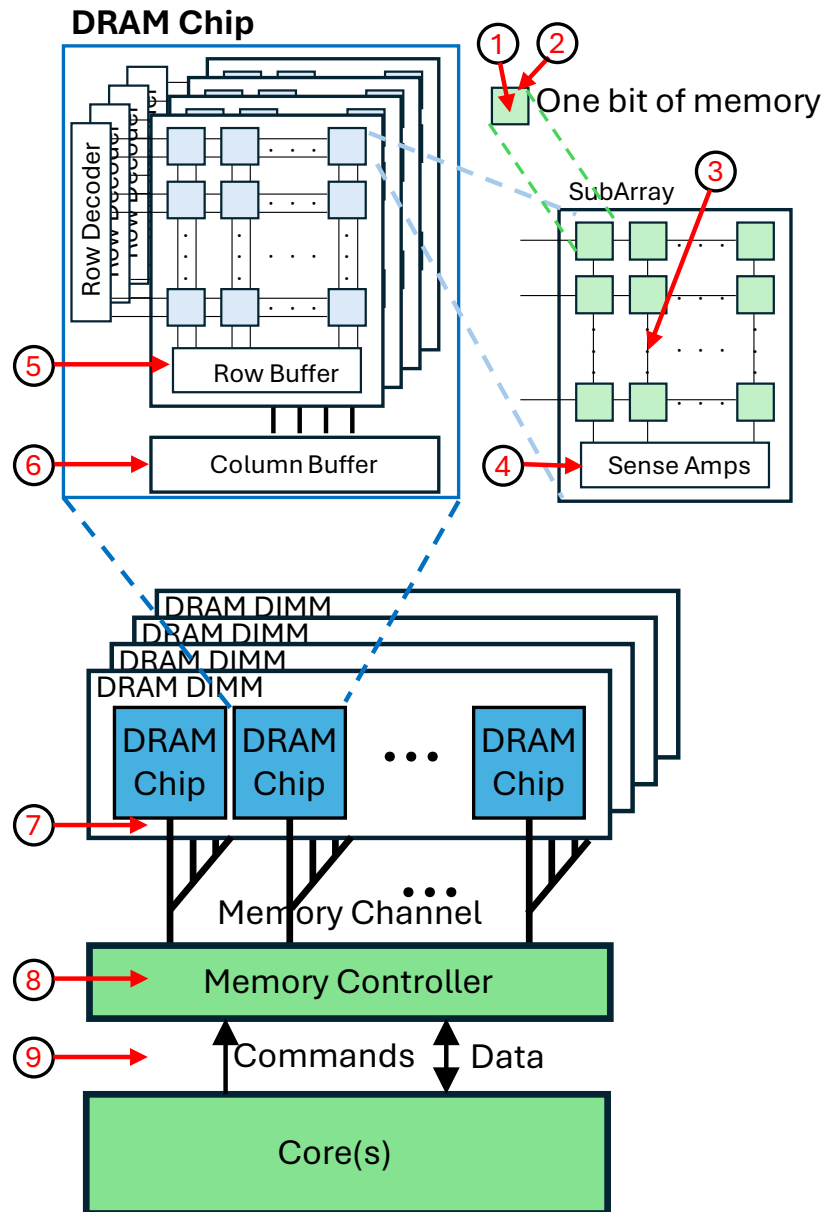
- Colossus 1: **7.4/13.6/300**
 - 150K H100, 50K H200, 30K GB200
- Colossus 2: **29.8/590/10,300**
 - 110K GB200
 - 440K GB300
- xAI Colossus: **38.6/876/14,700**
 - Last week: +220K GB300's
 - By Dec: another 220K

GPU	FP64	FP32	FP64 TC	FP32 TC	FP16 TC	FP8 TC	FP4 TC	INT8 TOPS	Memory GB	Memory B/W TB/s
H100 Hopper	30	60	60	1000	2000	4000		4K	80	3.35
H200	34	67	67	989	1979	3958		3959	141	4.8
GB200 Blackwell	40	80	40	1250	2500	6000	10K	5K	186	8
GB300 Ultra		80	1.3K*	2.5K*	5K*	10K*	20K*	330	279	8

Units of Exa

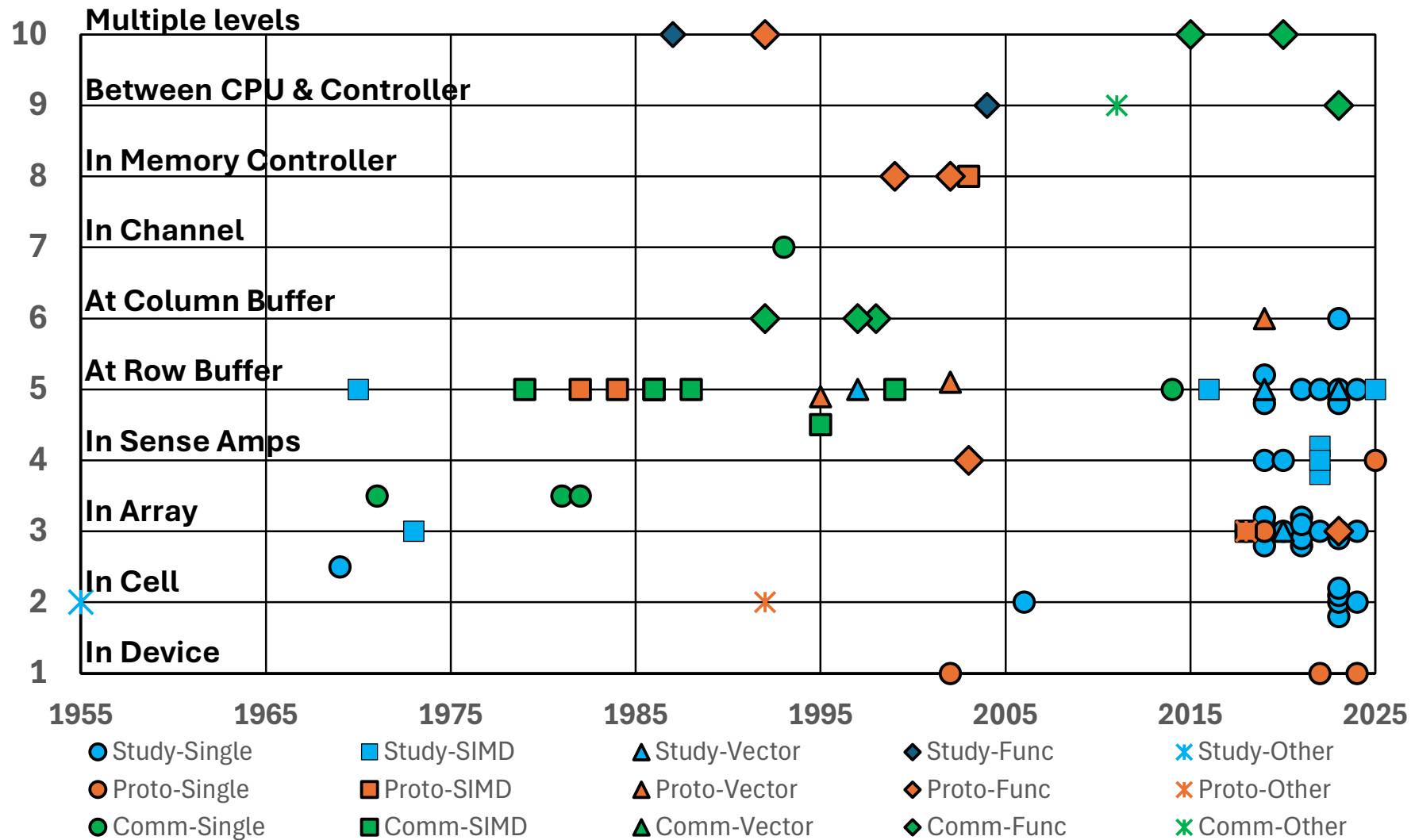
- FP64**
- FP64 TC**
- FP4 TC**

Attacking the Memory Bandwidth Issue - PIM

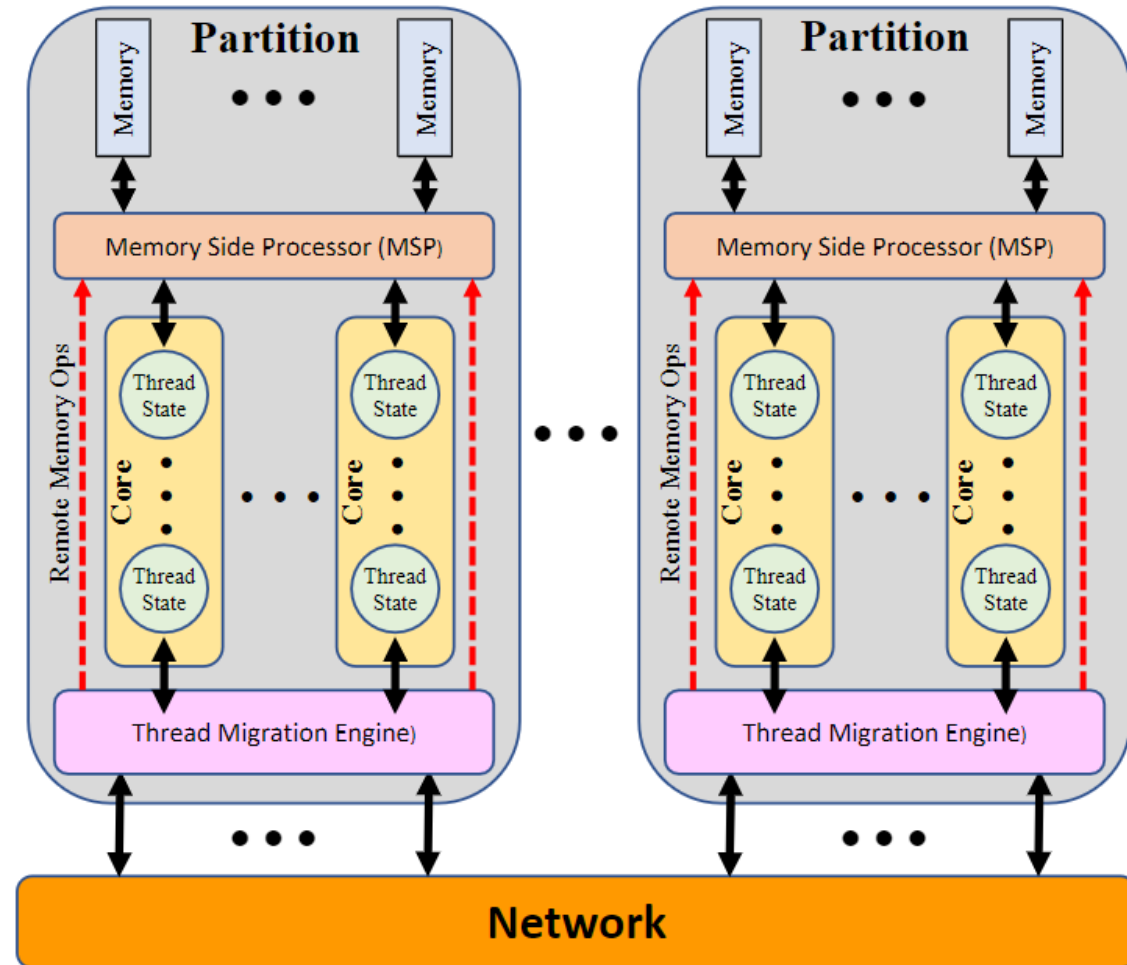


- PIM: Move processing in/near Memory
- Multiple opportunities
 1. In the cells that make up individual bits.
 2. Within a memory subarray, especially on the bit columns, but not in the cells.
 3. Within the sense amps of a subarray.
 4. At/in a buffer associated with a subarray or bank of subarrays.
 5. At/in a buffer associated with multiple banks.
 6. Between multiple memory chips and a memory controller.
 7. Within the memory controller.
 8. Between the processors and the memory controller.
 9. Combinations of the above, especially levels 2 through 5.

PIM Actually Has a Long History



How Do We Move the Computation to Memory?



- *Migratory Threading*
- All memory on a single logical space
- On nonlocal reference: H/W
 - Suspends thread
 - Packages State
 - Sends to correct node
 - Unpacks State
 - Restarts thread
- Cheap spawns



32 node Lucata Pathfinder
at GaTech's Rogue's Gallery

Conclusions

- 2008 Study nailed need for SIMD many-core, stacked memory, networks based on high radix switches
- But 2013 technology was insufficient
 - Too many endpoints, too much power lost to movement
- Frontier leveraged better technology
 - With wider SIMD, multi-die packaging, better networks & cooling
- More nuanced answer to “Did Frontier achieve exascale goals?”
 - **Yes** if flop-intensive
 - **Not** if memory or bandwidth-intensive
- Zettascale in 2036? AI Systems may be there now
- Memory Issues only getting worse
 - PIM may be an answer

Thank You!

Esp. Bill Harrod for all the Exascale studies
And to DOE for pushing to fruition